

# Ibrahim Khaliq

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## EDUCATION

**University of Richmond (UR), Richmond, VA**

May 2028

Bachelor of Science, Computer Science & Mathematics, Minor in Data Science and Statistics

**Relevant Courses:** Data Structures, Discrete Structures, Software Systems and Development, Linear Algebra, Computer Organization, Machine Learning, Fundamentals of Abstract Mathematics, Business Statistics

## COMPUTER SKILLS

**Languages:** Python, Java, C++, JavaScript, SQL, HTML/CSS, VB.NET

**Frameworks/Libraries:** FastAPI, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Streamlit

**Tools/Platforms:** Git/GitHub, Docker, AWS, Jupyter, Android Studio, Health Connect, Linux

**Concepts:** REST APIs, ETL Pipelines, End-to-End ML Pipelines, Data Processing, Testing, Q/A, Automation

## EXPERIENCE

**Undergraduate Research Assistant, SASSEMA (Psychology Study App), UR**

Jan 2026 – Present

- Led Android port of iOS ADHD ecological momentary assessment app in Java; built onboarding flow, day-2+ recovery logic, and dynamic rendering of 20+ questionnaires (200+ fields).
- Engineered background motion sensing (accelerometer + gyroscope, 25–50 Hz, duty-cycled) with AlarmManager-based reminders for reliable in-the-wild participant data capture.
- Built WiFi-gated batched JSONL upload pipeline (10KB payload threshold, multipart/form-data server sync) to eliminate cellular data cost; integrated Android Health Connect via Kotlin to Java interop for 5 passive data types.
- Diagnosed Android 13 Health Connect platform limitation and proposed hybrid Health Connect + SensorManager fix; authored 22-item code review, ran 45-case QA plan, and shipped debug APK.

**Resident Assistant (RA), Residence Life and Housing, UR**

Aug 2025 – Present

- Serve as mentor and first-response contact for 30+ residents, leading community programs with an 8-person RA team while handling conflict resolution, wellness concerns, policy issues, and referrals to campus resources.

**Innovation Fellow, Richmond Innovation Fellows (RIF), UR**

Jan 2025 – Present

- Selected as 1 of 8 fellows from 100 applicants; collaborated with founders, engineers, and venture capitalists on technology-driven solutions.
- Conducted 15+ stakeholder interviews to identify resource gaps in nonprofit food access, translating user insights into iterative solution prototypes.
- Evaluated 20+ databases and technology platforms to assess market need and technical feasibility for Sukoon, an AI-based mental health platform for South Asia.
- Synthesized user research, market analysis, and product strategy into pitches, advancing to the top 8 of the National Collegiate Innovation Challenge.

**SWE Intern, Express Tax and Accounting, Falls Church, VA**

Aug 2025 – Sep 2025

- Optimized SQL queries and added composite indexes across PostgreSQL/SQL Server tables, cutting financial report generation time by 40% and eliminating timeout errors on month-end close.
- Built Pydantic validation schemas and structured error-handling across financial reporting pipelines, eliminating 200+ monthly duplicate-entry errors flagged in prior audit cycles and cutting reconciliation time.
- Built Python/Pandas scripts to parse, reconcile, and flag mismatches across 5,000+ monthly ledger transactions from 3 source systems, cutting manual entry time by 35% and surfacing errors before quarterly close.
- Deployed an internal ETL pipeline that automated client tax document ingestion, parsing, and normalization into a central database, saving auditors 10+ hours weekly during peak filing seasons.

## PROJECTS

**AI GitHub Repository Assistant | Python, FastAPI, Docker, Ollama, GitPython**

- Built an AI-powered developer platform using Python, FastAPI, Docker, Ollama, and GitPython that clones and indexes repositories, generates LLM-based code summaries, and answers natural-language questions about codebase architecture.
- Implemented hybrid semantic retrieval, vector embeddings, parallel indexing, and commit-aware caching to improve retrieval quality and reduce redundant repository processing.

**Real-Time Facial Recognition System | Python, OpenCV, face\_recognition (dlib), NumPy**

- Built a real-time face detection and recognition pipeline in Python using OpenCV and dlib-based facial encodings to identify individuals from a live webcam feed at 15+ FPS on standard hardware.
- Automated timestamped attendance logging to CSV and designed an extensible identity-registration workflow enabling new faces to be added at runtime without code changes or model retraining.

**Real-Time Voice Assistant | Python, ElevenLabs Conversational AI SDK**

- Developed a real-time, voice-enabled AI assistant using the ElevenLabs Conversational AI SDK, driving natural spoken conversations through live microphone/speaker I/O with sub-500ms response latency.
- Engineered dynamic prompt/context injection and callback-based logging of agent responses, interruptions, and user transcripts, with secure credential management via environment variables.